

FHE + Neuromorphic Complete Reference Design

ENER / NeuroFHE Relay - revision B

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This package defines a physical relay board, its implementable digital circuits, and the matching patent architecture. An existing headset or acquisition device supplies digital telemetry. A Raspberry Pi 4 runs the local adapter, policy and FHE client; an ICE40UP5K implements the event and spiking circuits. A remote OpenFHE evaluator receives encrypted features and returns encrypted scores.

Deliverable	Contents
Patent schematics	16 A4 black-and-white vector sheets with matching descriptions and reference numerals.
Circuit schematics	8 A3 landscape sheets: four component sheets plus four digital/FHE datapath sheets.
Circuit design files	55 board components; 226 explicitly assigned pins; 32 named nets; BOM, pin map, SVG and legacy KiCad import sources.
Actual FPGA implementation	Synthesizable source, generated Verilog, fitted netlist, package constraints, routed configuration image and behavioral tests.
Specification working copy	New figure descriptions and a concrete hardware embodiment inserted into a separate copy of the original specification.

Evidence and limits

The actual encoder logic passed 74 simulated frames, 14,262 input samples and 4,736 counter comparisons. Place and route passes at 16 MHz; reported maximum frequency is 21.38 MHz. These are software verification results. The relay PCB, assembled electronics, headset-specific driver and distributed FHE deployment have not been physically validated.

The semiconductor choices and the eight-neuron dynamics are concrete design work in this revision. They are not represented as historical prototypes or measured clinical results. No patent has been filed or repository changes published.

ENER patent source, GitHub main

1. Architecture decisions

Decision	Reference choice and reason
Acquisition boundary	Use the original device and its digital USB interface or receiver. The relay does not add patient-connected analog electrodes or redesign the headset.
Trusted host	Raspberry Pi 4 Model B, 4 GB, 64-bit OS. Existing CPU, DDR, storage controller, network and power circuits remain within this off-the-shelf assembly.
Neuromorphic fabric	ICE40UP5K-SG48ITR50, 48-pin QFN plus exposed paddle. Clock is a 16 MHz external oscillator. No external FPGA memory or flash is required.
Configuration	Pi SPI slave configuration of FPGA SRAM. Same four SPI pads become the runtime serial link after configuration.
Encoder profiles	SPATIAL-64-B: repository-compatible event counter. LIF-64-B: concrete eight-neuron, frame-local integrate-and-fire network.
Feature contract	64 unsigned two-bit counts, $\text{index} = 8 * \text{timeBin} + \text{unit}$; 50 ms frame; eight equal time bins; eight units on a 4x2 map.
Homomorphic circuit	OpenFHE BFVRns scalar ciphertext/plaintext linear scoring. Two classes in the reference integration; arbitrary approved C is a software generalization.
Default disclosure policy	Encrypt all 64 feature positions, including zeros. Support-visible sparse export requires an explicit profile permission.
Output decision	Decrypt and perform signed range checks and argmax locally. The evaluator never receives the FHE secret key.

A complete CPU motherboard schematic is supplied by the board vendor; it is not newly designed here. The newly designed mezzanine is fully enumerated in the component drawings and connectivity files. This module boundary is part of the reference implementation, rather than an unspecified processor placeholder.

Raspberry Pi 4 Model B datasheet, release 1.1

Lattice iCE40 UltraPlus datasheet, FPGA-DS-02008 rev. 2.4

2. Acquisition and buffering contract

```
Canonical sample = (channel, timestamp_us, amplitude)
channel           : uint8, allowed 0..7
timestamp_us      : uint16, allowed 0..49999 within a frame
amplitude         : int16, calibrated integer units
frame            : interval [origin, origin + 50000 us)
unit_x            : channel mod 4
unit_y            : floor(channel / 4)
```

The device adapter converts the original SDK or transport format into this contract. It must bind channel order, sampling rate, scale, offset, polarity and calibration version to the encoder/model profile. The hardware default threshold of 50 is in calibrated integer units; it is not automatically 50 microvolts. The exact headset model has not been supplied, so this package defines the required digital interface rather than claiming an untested vendor driver.

Deterministic preparation

Use device timestamps or a validated sample-counter clock. Reject missing channels, unknown channels, non-finite values, amplitude conversion overflow and device sequence gaps. Assign each sample to the half-open 50 ms interval. Stable-sort each complete frame by timestamp, preserving original order for ties. Do not substitute arrival order for device time. Retain a frame validity flag that prevents any invalid frame from being encrypted.

Buffer	Capacity and ownership
Raw frame bank A/B	Two host buffers, each 512 records x 8 bytes = 4096 bytes. One acquisition writer and one encoder reader; swap only complete valid frames.
Input rate envelope	Up to eight channels x 1000 samples/s for the electrical budget: about 400 records per 50 ms. The exact device/driver rate still requires measurement.
FPGA feature store	One 64 x 2-bit frozen/active store. BEGIN clears; END freezes. Read all 64 positions before issuing the next BEGIN.
HE work queue	Eight feature records maximum; each holds counts plus immutable frame/profile metadata. No unbounded backlog.
Overrun	Invalidate the affected acquisition frame. If HE queue is full, defer or use an approved local path; log a gap. Never relabel an old frame as current.

Compression occurs before FHE encryption. The local host sees raw samples because it is inside the chosen trusted boundary. This is not a design in which plaintext remains secret from the local operating system.

3. Host and FPGA pin assignments

Signal	Pi physical / BCM	FPGA pad	Path / behavior
SCK	23 / GPIO11	15	U8 -> R1; mode-0 clock
MOSI	19 / GPIO10	17	U8 -> R2; host to FPGA
MISO	21 / GPIO9	14	R3; FPGA to host
CS_N	24 / GPIO8	16	U9 -> R4; exactly 64 clocks per runtime transaction
CRESET_N	11 / GPIO17	8	Open-drain host control, U7 and SW1; external 10 kohm pull-up
CDONE	13 / GPIO27	7	Configuration status; keep pin reserved for configuration
READY	15 / GPIO22	9	High permits starting a command; falls during core/reply work
FRAME_READY	16 / GPIO23	10	High only for a frozen valid frame
SOFT_RESET	18 / GPIO24	12	U9 -> R6; synchronized into logic
Domain rst	Not exposed	11	Tied to GND; FPGA initialization plus synchronized soft reset is used
Clock	Not exposed	35	Y1 -> R5; 16 MHz global clock
Power	2,4 / 5 V	Supply unit	F1, U2-U7 sequenced rails
Ground	6,9,14,20,25,30,34,39	49 / paddle	Continuous ground plane

J1 pin numbers are physical connector numbers, not BCM GPIO numbers. Pins 1 and 17 are deliberately not joined to the mezzanine 3.3 V regulator. SCK, MOSI, CS and soft reset pass through partial-power-down-capable buffers so a powered host does not directly drive those FPGA input pads when their local rail is off. R12-R14 and R18 define the buffer input levels.

Raspberry Pi 4 Model B datasheet, release 1.1

Lattice UltraPlus breakout-board schematic, Appendix A, rev. 1.2

Texas Instruments SN74LVC2G17 datasheet, rev. N

4. Power tree and supervisor design

Rail / part	Load allocation	Control
+5V_RAW -> F1	150 mA relay input allocation	0.25 A hold PTC; reference 5 V operating window 4.75..5.25 V
U2, TLV75512P	1.2 V / 100 mA core+PLL allocation	EN tied to +5V; first rail
U5, TPS3808G12	Sense 1.2 V; nominal trip 1.12 V	RESET_N pulls IO_ENABLE low until threshold + CT delay
U3, TLV75533P	3.3 V / 30 mA I/O, clock, buffers	EN=IO_ENABLE; does not depend on Pi 3.3 V rail
U6, TPS3808G33	Sense 3.3 V; nominal trip 3.07 V	RESET_N controls VPP_ENABLE
U4, TLV75525P	2.5 V / 10 mA configuration allocation	EN=VPP_ENABLE
U7, TPS3808G25	Sense 2.5 V; nominal trip 2.33 V	RESET_N holds CRESET_N low until last rail is valid

Each TPS3808 receives +5V on pin 6, ground on pin 2, manual-reset high on pin 3 and a 1 nF C0G capacitor from pin 4 to ground. Pin 5 senses its assigned rail. Pin 1 is an open-drain output. Nominal delay is $CT[nF]/175 + 0.0005$ seconds = 6.21 ms. Delays are illustrative nominal component behavior, not measured board startup.

U2-U4 each have a local input capacitor and a 10 uF output capacitor. Effective output capacitance must remain at least 1 uF after DC-bias and tolerance derating. C14-C19 sit at the respective FPGA supply pins. R17 with C20/C21 filters VCCPLL. The ground paddle must be soldered and tied into the plane.

Thermal sizing

At 5.25 V input and the 100 mA core allocation, U2 dissipates approximately $(5.25-1.2)*0.1 = 0.405$ W. Using the listed DBV junction-to-ambient value of 100.8 C/W gives a first-order rise of about 41 C. A 50 C local ambient would imply about 91 C junction before board-specific corrections. This is an allocation check, not a measured FPGA current or a substitute for layout thermal analysis. Verify rail ramps, output load and regulator temperatures on the assembled board.

Texas Instruments TLV755P datasheet, rev. D

Texas Instruments TPS3808 datasheet, rev. N

Lattice iCE40 UltraPlus datasheet, FPGA-DS-02008 rev. 2.4

5. Clock, reset and configuration sequence

Y1 is SiT8008BI-12-33E-16.000000: 2.5 x 2.0 mm, 3.3 V, industrial temperature range, nominal 16 MHz, 25 ppm stability, output-enable option. Pins are 1=OE, 2=GND, 3=OUT, 4=VDD. OE is tied high; C22 bypasses VDD. R5 is placed near the source. FPGA pad 35 accepts the global clock. No PLL multiplication is used.

Power-up / configuration procedure

1. Keep host signal GPIOs non-driving until local rails have settled; keep GPIO17 low or high-impedance only. Wait at least 100 ms after board power before initiating configuration.
2. Reserve SPI0 exclusively. Pull CRESET_N low and drive CS_N low to select the FPGA slave configuration mode. Hold reset at least 1 ms.
3. Release CRESET_N to the external pull-up; confirm CDONE is low. Wait at least 1200 us for configuration-memory clearing, following the Lattice slave procedure.
4. Raise CS_N and issue eight dummy clocks. Then lower CS_N and send the complete bitstream MSB first at 1 MHz, with configuration data changing on falling clock edges.
5. After the complete image, raise CS_N; wait/check CDONE and provide at least 49 additional dummy clock cycles. Reject a configuration that never raises CDONE.
6. Change to the runtime mode-0 protocol, assert then release SOFT_RESET for at least 10 us, and query NOP/version. Require reply 0x0201 and zero errors before BEGIN.

A CRC error, reset, CDONE loss or invalid frame requires a new BEGIN and a fresh frame. The 64-count store is volatile. SRAM configuration and the local secret key are not recovered from old network ciphertexts.

SiTime SiT8008B datasheet, rev. 1.08

Lattice iCE40 programming and configuration, section 13

6. Runtime serial command format

SPI is mode 0, MSB first, at no more than 1 MHz. A transaction is exactly eight bytes while CS_N is low. Require at least 1 us from CS assertion to first rising clock, at least 1 us after the last clock before CS release, and READY high before asserting the next CS. A native host driver must meet these electrical timings; a generic Linux/Python call is not a real-time throughput guarantee.

Byte	Field	Encoding
0	Opcode	Command below
1	Argument	Mode, channel or feature index
2..3	Timestamp	Unsigned big-endian microseconds within frame
4..5	Value	Signed big-endian int16 for samples; unsigned bit pattern for configuration
6	Sequence	Host increments modulo 256; echoed in the next response
7	CRC-8	Polynomial 0x07, initial 0, no reflection, no final XOR; covers bytes 0..6

Opcode	Command	Action / legal state
0x00	NOP / VERSION	Any state when READY; payload 0x0201
0x01	BEGIN	Argument 0=SPATIAL-64-B, 1=LIF-64-B; clears count/state/error latch; supersedes an abandoned frame
0x02	END	Only an active frame; freezes only if frame has no fault
0x10	SAMPLE	Argument=channel 0..7; time <50000; signed value; active frame only
0x11	READ_COUNT	Argument=index 0..63; frozen valid frame only
0x20	SET_THRESHOLD	Value 1..32768, outside active frame; clears frozen-frame validity
0x21	SET_THETA	Value 1..65535, outside active frame; clears frozen-frame validity

The one-command response pipeline permits each outgoing request to clock out the previous response. Issue a final NOP to retrieve the final READ_COUNT reply. No SPI command writes a cryptographic key, model score or external-network destination.

7. Replies, fault flags and recovery

Byte	Meaning
0	Status: bit 0=frame active; bit 1=frozen frame valid; bits 2..7=0
1	Echoed opcode
2	Echoed request sequence
3	Echoed argument
4..5	Unsigned big-endian payload; READ_COUNT returns 0..3
6	Sticky error flags
7	CRC-8 over response bytes 0..6

Bit	Cause	Required host response
0	Wrong length, nonzero CRC residue or command while unavailable	Discard current frame; synchronize replies, then BEGIN a new frame
1	BEGIN mode not in {0,1}	Reject profile selection
2	END without active frame	Repair host lifecycle; do not use resulting status alone
3	Invalid channel/time or decreasing timestamp	Discard frame; investigate acquisition/order mapping
4	READ without valid frozen frame or index >=64	Discard readout
5	Invalid configuration or configuration while frame active	Discard frame; configure only between frames
6	Unknown opcode	Reject protocol-version mismatch
7	Reserved	Treat nonzero as unsupported protocol

The host checks response CRC, opcode, sequence, argument, status and error byte. Sequence echo is a local transport association check, not an authentication mechanism. BEGIN deliberately clears previous errors, so the host must latch any error before beginning again. A failed or corrupt frame is never silently converted into a valid all-zero feature vector.

```
BEGIN(profile)
for sample in stable_sorted_valid_frame:
    wait READY; SAMPLE(channel, time, amplitude)
wait READY; END
require FRAME_READY and zero error flags
for index in 0..63: READ_COUNT(index)
NOP # clocks out the last read response
verify every response; hand counts to policy/FHE
```


8. Spatial counter circuit

SPATIAL-64-B realizes the deterministic sorter in prototype/lib/spike-sorter.mjs. The host supplies canonical channels in stable timestamp order; the FPGA implements magnitude thresholding, per-channel refractory acceptance, time-bin addressing and bounded counting. Spatial mapping is $x = \text{channel} \bmod 4$, $y = \text{floor}(\text{channel}/4)$. It does not perform waveform clustering or claim to identify individual biological neurons.

```
b = floor(timestamp_us / 6250)
i = 8*b + channel
candidate = abs_int17(amplitude) >= threshold
refractory_ok = !seen[channel] OR
               timestamp_us - last[channel] >= 100
if candidate AND refractory_ok AND count[i] < 3:
    count[i] = count[i] + 1
    last[channel] = timestamp_us
    seen[channel] = 1
```

The magnitude path is 17 bits so -32768 is represented as magnitude 32768 rather than overflowing a signed 16-bit absolute-value circuit. Eight time bins are derived by threshold comparisons; there is no variable divider. Both 0 us and 49999 us are legal. A timestamp of 50000 belongs to the next frame. Exactly 100 us after the last acceptance is legal. At saturation, the count and last-accepted timestamp both remain unchanged, matching the source sorter.

State element	Width / behavior
Counts	64 x 2 bits = 128 bits; register array
Last accepted time	8 x 16 bits = 128 bits
Seen flags	8 bits
Frame/order state	Previous 16-bit timestamp, valid flag, active flag and fault latch
Clear engine	6-bit address, one counter cleared per clock
Output	READ_COUNT zero-extends the two-bit value to 16 bits

All state is local to a frame and cleared on BEGIN. A continuous cross-frame refractory or recurrent-state implementation would be a different profile and is not silently substituted here.

Repository sorter source

9. Concrete integrate-and-fire network

LIF-64-B uses the same eight canonical input channels, threshold gate and 100 us source-event refractory interval. Each accepted source event visits eight output neurons in order 0..7. The fixed synaptic matrix is below; rows are output neurons and columns are input channels. Values are dimensionless integer integration increments.

n / c	0	1	2	3	4	5	6	7
0	16	0	0	0	0	0	0	4
1	4	16	0	0	0	0	0	0
2	0	4	16	0	0	0	0	0
3	0	0	4	16	0	0	0	0
4	0	0	0	4	16	0	0	0
5	0	0	0	0	4	16	0	0
6	0	0	0	0	0	4	16	0
7	0	0	0	0	0	0	4	16

Before any event at time t:

advance each missing 1 ms tick in ascending order

for every neuron n:

V[n] = V[n] - (V[n] >> 4)

R[n] = max(0, R[n] - 1)

For an accepted event on channel c:

for n = 0..7:

if W[n,c] != 0 and R[n] == 0:

u = V[n] + W[n,c] # 17-bit sum

if u >= theta:

V[n] = 0; R[n] = 2

count[8*floor(t/6250)+n] = min(3, count[...] + 1)

else: V[n] = u

V is an unsigned 16-bit register; R is a two-bit tick counter; theta defaults to 64. Refractory duration is two tick advances, so the elapsed interval depends on event phase within the millisecond; it is not an exact two-millisecond timer. Increments with weight zero have no effect. A neuron resets on a spike even if its output count is already saturated. There is no cross-frame state, recurrence through output spikes, inhibitory weight or online training in this selected network.

10. Local controller and memory schedule

State	Operation	Bound
IDLE	Admit one checked command; latch profile or sample	One synchronous admission
CLEAR	Clear one of 64 count registers per clock	64 clocks after BEGIN admission
ADVANCE	Apply missing 1 ms leak ticks to all eight neurons	At most 49 ticks in a 50 ms frame
CHECK	Apply amplitude/refractory gates; spatial update or dispatch synapses	One decision cycle
SYNAPSE	Select neuron, ROM weight and state; integrate/reset/count	8 clocks per accepted source event
REPLY CRC	Finalize 56-bit response body with serial CRC	57 clocks; READY stays low

The measured maximum core command latency in the supplied simulation is 65 clock cycles. A conservative protocol budget permits 128 clocks (8 us at 16 MHz) between the final CS release and the next transaction, while still requiring READY. Serial shifting and input synchronizers use the same clock domain. No asynchronous FIFO or unsynchronized multi-bit bus is present.

Memory / storage	Concrete allocation
FPGA	Count registers, timestamp registers, neuron registers, transport shift registers, CRC state and FSMs. ROM weights are fixed constants synthesized into logic.
Local host	Two raw-frame banks; eight-entry feature queue; public model; FHE context; active keys; bounded job/reply ledger.
Remote worker	Public model/context/key, up to 64 input ciphertexts, two encrypted accumulators and one product temporary.
Persistent state	Versioned public profiles and signed application binaries. Secret persistence is optional and outside FPGA hardware; volatile session keys are the selected default.

The source code is the editable behavioral hardware definition. ener_relay.v is generated, and ener_relay.json / ener_relay.asc / ener_relay.bin are synthesis, routed-design and configuration artifacts. The binary is a build result for this exact pinout, not a claim that a physical board has been programmed.

11. Privacy representation and model binding

Profile	Export	Consequences
DENSE64, default	64 independently randomized ciphertexts in fixed index order; zeros included	Hides active support in the message contents under HE assumptions. Transport timing and profile identifiers remain visible.
SPARSE64, explicit opt-in	Strictly increasing active indices and ciphertexts of their values	Reveals exact support and active count. With binary counts, revealed support determines the feature vector.
Other embodiments	Padded/coarsened/encrypted-index approaches remain disclosed alternatives	Not implemented by simply relabeling this sparse kernel. Encrypted indices require an oblivious selection circuit.

Each approved model includes encoder ID, calibration digest, mode, threshold, theta, frame duration, shape [8,8], channel-map digest, count bound, integer weights, integer biases, class order, representation policy and HE-context digest. A model trained for SPATIAL-64-B is not valid for LIF-64-B merely because both have 64 outputs. A profile update takes effect only between frames and invalidates pending incompatible readouts.

Integer score range

```

y[k] = b[k] + sum(i=0..63, W[k,i] * x[i])
x[i] in {0,1,2,3}; t = 65537
B[k] = abs(b[k]) + 3 * sum_i abs(W[k,i])
Require max_k B[k] <= 32768

```

For a convenient bounded model family, $\text{abs}(W[k,i]) \leq 127$ and $\text{abs}(b[k]) \leq 8191$ imply $B[k] \leq 32575$. Center-lift a decrypted residue r by returning r for $r \leq 32768$ and $r-65537$ otherwise. This prevents ambiguity from plaintext modular wrap. It does not by itself prove adequate ciphertext-noise margin.

12. FHE context and evaluation circuit

The baseline follows the repository native OpenFHE BFVRns linear scorer. It uses scalar feature ciphertexts and a public integer model. Key generation and decryption occur locally; remote computation is ciphertext/plaintext multiplication and ciphertext addition. The selected circuit needs no bootstrapping, rotation or ciphertext/ciphertext multiplication.

```
CCParams<CryptoContextBFVRNS> p;  
p.SetPlaintextModulus(65537);  
p.SetMultiplicativeDepth(1);  
p.SetBatchSize(1);  
p.SetSecurityLevel(HEStd_128_classic);  
cc = GenCryptoContext(p);  
cc->Enable(PKE); cc->Enable(LEVELEDSHE);  
keys = cc->KeyGen();  
  
for each exported i:  
    c[i] = cc->Encrypt(pk, cc->MakePackedPlaintext({x[i]}));  
for each class k:  
    Y[k] = cc->Encrypt(pk, cc->MakePackedPlaintext({b[k]}));  
    for each exported i:  
        term = cc->EvalMult(c[i],  
                             cc->MakePackedPlaintext({w[k,i]}));  
        Y[k] = cc->EvalAdd(Y[k], term);
```

Use OpenFHE parameter generation to select N and the CRT primes, then record the actual N, each q_j , their total modulus bits, library version, build configuration, security setting and serialization digest in the public context manifest. Fixed parameters mean a fixed approved serialized context per deployment; they do not mean inventing unverified primes. Fail closed if the approved context differs from the received context.

In the reviewed BFVRns parameter generator, multiplicativeDepth, evalAddCount and keySwitchCount cannot be set nonzero together. Therefore this design does not combine depth=1 with evalAddCount=64. Validate the actual worst-case linear circuit and its maximum approved weights with native round-trip tests before admitting a deployment; do not treat a depth field as a complete noise proof.

[OpenFHE BFVRns parameter-generation source](#)

[OpenFHE example parameter guidance](#)

13. Host and remote process interface

Boundary	Required contract
Acquisition -> encoder	Canonical frame with immutable origin, sample-clock version, stable order and validity. Only local IPC / memory.
Encoder -> HE queue	64 counts; frame ID; encoder/config digest; calibration/channel-map digest; validity; no raw waveform copy.
Client -> evaluator	Protocol version; random 128-bit job ID; monotonic session counter; context/key/model/profile digests; shape; representation; bounded ciphertext blob lengths and payload.
Evaluator -> client	Same job and digest bindings; ordered C ciphertext scores; server protocol version; explicit error code for public admission failures.
Client -> application	Locally decrypted signed scores and approved class result; frame timestamp; validity/freshness; provenance digest.

Use an authenticated encrypted channel with server identity validation. Serialize an explicit length-prefixed public header followed by individually length-prefixed OpenFHE binary ciphertexts; reject oversized lengths before allocation. Limit a reference request to 64 ciphertexts and a response to two. Set a deployment-specific byte cap after measuring the approved context; do not infer a ciphertext byte size solely from plaintext dimensions.

The worker validates version, model/context/key/profile IDs, class count, feature domain, absence of duplicate indices and the declared representation before evaluation. For DENSE64, require every index 0..63 exactly once in order. For SPARSE64, use only the disclosed indices and reject repetitions or out-of-range values. Neither client nor worker should deserialize arbitrary objects outside the approved schema.

The client binds every response to one outstanding job, accepts it once, rejects expired or mismatched jobs, and performs decryption only after public checks. Correct execution by an actively malicious worker is not proven by transport authentication; a separate verifiable-computation protocol or trusted evaluator would be needed for that stronger property. Do not expose detailed local decryption failures as an oracle to the worker.

This is the completed process/interface design. The repository native demo co-locates client and evaluator; it does not demonstrate this network deployment or hardware integration.

14. Key lifecycle and local decision

Phase	Required action
Session start	Create context and keypair locally from cryptographic OS entropy; log only public identifiers. No deterministic demo seed in production key generation.
Provisioning	Send public key and context. Send evaluation keys only for operators that require them; this linear circuit does not require rotation or relinearization keys.
Queueing	Attach immutable context/key/model/encoder IDs; protect raw/feature buffers from untrusted local processes according to the deployment trust model.
Rotation	Stop admitting jobs under the old key, drain or expire their bounded ledger, distribute the new public context/key and resume.
Result use	Check bindings, decrypt locally, validate signed score range, apply the approved tie-break and output policy.
Session end	Stop transmission, invalidate outstanding jobs and release secret-bearing memory. For a strong erasure requirement, use an implementation with explicit zeroization and verify memory/swap behavior.

Reference tie-break: choose the first class in the signed model manifest when scores tie. A timeout or invalid response produces no new classification. Retain the last result only if the application explicitly labels it stale. The reference relay returns an informational result; it does not directly stimulate neural tissue or drive a safety-critical actuator.

The local host and its operating system are trusted for plaintext and secret-key handling. No claim is made that commodity Pi RAM is a hardware secure enclave. The FPGA has no key register or key-loading command. The remote worker sees public model weights and allowed metadata; model confidentiality is not provided by this public-weight embodiment.

15. Throughput and capacity budget

Quantity	Calculation / result
Core clock	16 MHz = 62.5 ns per cycle
Runtime SPI	1 MHz = 64 us for an eight-byte transaction
Timing budget per command	64 us clocking + 1 us setup + 1 us hold + 8 us recovery = 74 us
Upper input budget	8 channels x 1000 samples/s x 0.05 s = 400 SAMPLE commands
Frame transaction count	BEGIN + 400 SAMPLE + END + 64 READ + final NOP = 467
Wire/logic budget	467 x 74 us = 34.558 ms per 50 ms frame, before host scheduling overhead
250 samples/s/channel	Average 100 samples/frame: about 12.358 ms at the same conservative transaction budget
FPGA storage	Count store 128 bits; neuron potentials 128 bits; timestamps 128 bits; plus flags, counters, transport and control state
FHE input count	Dense: 64 ciphertexts. Sparse: A, 0..64, and disclosed support.
FHE kernel work, C=2	Dense: 128 CT/PT multiplies + 128 CT adds, two bias encryptions and two score decryptions. Sparse: 2A multiplies/adds.

These are explicit design budgets, not measured end-to-end rates. Host GPIO/syscall scheduling, device decoding, FHE encryption, networking and decryption must be measured on the target. Admission control uses measured context-specific service times. A scalar sparse baseline does not establish superiority over packed dense FHE; packing changes both arithmetic counts and communication cost.

The existing synthetic repository case has 18 active positions out of 64, giving 36 multiplies and 36 additions for two classes. Its plaintext expected scores are 9 and 51 for the repository demo model. Those fixtures establish a regression target, not accuracy on a real headset.

16. PCB implementation and commissioning

Implement a four-layer mezzanine with continuous ground reference. Follow the vendor SG48 land pattern and exposed-pad requirements. Keep each supply capacitor close to its named U1 pin; use short local return paths. Route the clock from Y1 through R5 to pad 35 with a continuous reference plane. Place R1/R2/R4 near the buffer outputs and R3 near the FPGA output. Keep header orientation unambiguous and provide a pin-1 marker.

Separate the 5 V feed and core-current path from the clock route. Use sufficient copper for the U2 thermal allocation. Do not connect the independent 3.3 V regulator to Pi header pins 1 or 17. The original headset interface stays within its vendor-approved digital connection. Mechanical clearance to Pi connectors and heatsink must be checked against the vendor drawing before PCB release.

Ordered bring-up checks

1. Before fitting the Pi or U1, inspect every power net for shorts and verify component package/pad orientation against the datasheet.
2. Use a current-limited bench source. Observe 1.2 V, then 3.3 V, then 2.5 V and CRESET_N with an oscilloscope. Verify monotonic ramps against Lattice limits; do not infer ramp compliance from the nominal LDO startup time.
3. Verify 16 MHz at Y1 and U1 pad 35, DC rail levels, power draw and temperature under a high-switching test image.
4. Configure the FPGA, verify CDONE, exercise the NOP/version command, and capture SCK/CS/MOSI/MISO/READY on a logic analyzer.
5. Replay the supplied synthetic and boundary fixtures; compare every returned count to the software reference. Test malformed packets, reset and acquisition loss.
6. Run native FHE round trips for zero, saturated, random and model-boundary feature vectors; verify exact signed scores for each approved context/model.
7. Integrate the actual headset adapter, then measure acquisition, encoding, encryption, network and result latency separately. Admit only the measured operating envelope.

The package stops at schematic, circuit implementation and software verification. There is no routed PCB, fabrication output, assembled-board measurement or claim of electrical/clinical qualification. The legacy KiCad files are import sources; their import and ERC have not been exercised in this environment.

17. Verification evidence

Check	Observed result
Behavioral core simulation	74 frames; 14,262 samples; 4,736 exact counter comparisons; both profiles.
Boundary cases	Empty frame, signed minimum, threshold equality, refractory equality, last valid timestamp, saturation and long gaps.
Invalid frame handling	Invalid channel, out-of-frame time, decreasing timestamp, injected link fault and active-frame configuration reject.
Serial circuit simulation	18 complete transactions at 1 MHz; bad CRC, short and long packets rejected; reset recovery checked.
Synthesis	Yosys synth_ice40 completed; CHECK found zero problems. 2013 LUT4 primitives, 887 flip-flops, 387 carry primitives.
Place and route	UP5K SG48; all nine user ports constrained. 2686 / 5280 logic cells. 16 MHz target passes; reported fmax 21.38 MHz.
Timing margin	62.5 ns target period versus approximately 46.78 ns critical path; about 15.72 ns margin in this tool report.
Board connectivity	55 components, 226 physical pins and 32 named nets; every pin drawn once; all named nets have at least two terminals.

Static timing covers the implemented internal synchronous logic. It does not establish asynchronous metastability probability, board-level signal integrity or external device timing; those are addressed by the specified synchronizers, interface budget and hardware qualification. The 16 MHz selection is based on the final routed result, not on an assumed device maximum.

Reproduce with the pinned tool versions in requirements-hdl.txt and the commands in README.md. The verification directory contains the raw logs, JSON reports and a serial waveform. Generated Verilog is an export of the same hardware model used in simulation.

[Amaranth simulator documentation](#)

[Yosys and nextpnr framework paper](#)

18. Patent drawing descriptions (1-8)

Figure 1 illustrates an acquisition device 100 and a trusted local relay 200 comprising normalization and ordering 210, a neuromorphic circuit 220, compressed feature memory 230, and policy-controlled FHE encryption 240. The relay provides encrypted features to a remote encrypted inference engine 300.

Figure 2 illustrates a physical embodiment in which a digital device interface 110 connects to a local host processor 250. The host exchanges commands and features with programmable neuromorphic logic 220 over a serial link 260. A sequenced power and clock subsystem 270 supplies the logic.

Figure 3 illustrates canonical sample formation 211, frame validation and stable ordering 212, amplitude/refractory gating 221, and selection 222 between a spatial counter 223 and a spiking-neuron encoder 224. Both profiles write time/unit counts to memory 230.

Figure 4 illustrates a signed sample register 401, an absolute-value circuit 402, and threshold comparator 403. Last-accepted timestamp memory 404 and a time-difference comparator 405 provide a refractory condition to acceptance logic 406.

Figure 5 illustrates time-bin comparators 501 and a spatial-unit address 502 feeding address formation 503. Bounded memory 230 is updated by a saturating increment circuit 504 and exposed through a frozen-frame readout 231.

Figure 6 illustrates selected connections in an eight-neuron local encoder 224. Each input drives a corresponding self-input synapse 601 and an adjacent-neuron synapse 602; the pattern repeats through unit seven and wraps to unit zero.

Figure 7 illustrates synaptic weight ROM 701, neuron state register 702, integration adder 703, threshold comparator 704, and spike/reset selector 705. A threshold crossing resets state and updates bounded output-count memory 230.

Figure 8 illustrates timestamp register 801, tick advance logic 802, shift-and-subtract decay circuit 803, and refractory down-counter 804. Tick updates occur before same-time input events and frame-local state resets on BEGIN.

18. Patent drawing descriptions (9-16)

Figure 9 illustrates synchronizer registers 901, edge detection and a serial shift register 902, length/CRC admission 903, command/frame controller 904, and reply/CRC circuit 905. Rejected serial frames invalidate the active feature window.

Figure 10 illustrates core supply 1001, core supervisor and I/O enable 1002, I/O supply/supervisor 1003, configuration supply 1004, and reset supervisor 1005. The logic is released after the required rails become valid.

Figure 11 illustrates application of an approved privacy profile 241 to local counts 230. A fixed-domain encrypted representation 242 or a permitted support-visible representation 243 is bound to model, context and layout metadata 244.

Figure 12 illustrates encrypted feature value 1201 and public model weight 1202 entering ciphertext/plaintext multiplication 1203. Product ciphertexts enter encrypted accumulator 1204, initialized from encrypted bias 1205.

Figure 13 illustrates local secret key 1301, public key/context 1302, local decryption 1303, remote public-key encrypted evaluation 1304, and authorized local result 1305. The secret key remains inside trusted domain 200.

Figure 14 illustrates a frame lifecycle comprising clear 1401, ordered admission 1402, local state update 1403, valid-frame freezing 1404 and read/encrypt/dispatch 1405. Fault latch 1406 prevents dispatch of an invalid frame.

Figure 15 illustrates task/privacy constraints 1501 and resource/quality measurements 1502 selecting among an approved encoder/model catalog 1503. Compatibility and range checks 1504 determine admission to a remote, local or deferred route 1505.

Figure 16 illustrates an optional training embodiment with governed data 1601, candidate encoder 1602, task-utility test 1603, reconstruction/identity attack 1604, held-out acceptance gate 1605, and frozen deployment profile 1606. The fixed reference spiking circuit does not claim to have passed this training process.

19. Disclosure coverage and implementation status

Existing disclosure	Figures / implementation	Status in this revision
F.1-F.4; J.1: local compression then encryption	Figures 1-5, 11-13; component C01-C04 and datapath C05/C08	Fully specified reference partition and interfaces
F.3.1; J.3/J.9: spatial sparse BFV	Figures 3-5, 11-12; SPATIAL-64-B	Actual hardware logic simulated; native software kernel already exists in source
F.3; F.13: neuromorphic hardware	Figures 6-10; LIF-64-B; U1 circuitry	New concrete circuit design; simulated and placed/routed; not physically built
F.5, F.11: adaptive profiles and routing	Figure 15; model/profile admission contract	Architecture specified; scheduler and live metrics are integration work
F.8, J.10: reconstruction-resistant training	Figure 16	Optional disclosed training architecture; not a claimed property of the fixed ring encoder
F.9-F.10: metadata and trust separation	Figures 11 and 13; explicit DENSE64/SPARSE64	Exposure defined; dense default is a new selected implementation policy
F.12: policy, audit and output	Figures 13-15; frame/job lifecycle	Local acceptance and network contracts specified
F.6-F.7; H: CKKS, TFHE, MPC alternatives	Retained in source specification	Not mixed into the selected BFVrns board/runtime path; separate profiles/circuits required

The reference circuit is one embodiment of the existing architecture. Standard LIF dynamics, ordinary power supervisors and conventional FHE operators are not individually asserted to be novel. This engineering document is not a novelty search, an inventorship determination or an allowance opinion.

The companion specification working copy replaces the six old brief drawing descriptions with the coordinated sixteen-figure set and inserts section F.14. Other original specification language is preserved. The original files remain unchanged. The working copy is intended for the inventor and patent professional to review with the final figures.

20. Bill of materials (1-19)

Ref	Value	Part / order specification	Package
J1	PI 40-PIN SOCKET	Specified generic	2x20, 2.54 mm, female, keyed orientation
U1	ICE40UP5K-SG48	ICE40UP5K-SG48ITR50	QFN-48, 7x7 mm, 0.5 mm pitch + exposed pad
U2	TLV75512P	TLV75512PDBVR	SOT-23-5 / DBV
U3	TLV75533P	TLV75533PDBVR	SOT-23-5 / DBV
U4	TLV75525P	TLV75525PDBVR	SOT-23-5 / DBV
U5	TPS3808G12	TPS3808G12DBVR	SOT-23-6 / DBV
U6	TPS3808G33	TPS3808G33DBVR	SOT-23-6 / DBV
U7	TPS3808G25	TPS3808G25DBVR	SOT-23-6 / DBV
Y1	SiT8008B / 16 MHz	SiT8008BI-12-33E-16.000000	2520, four pads
U8	SN74LVC2G17	SN74LVC2G17DBVR	SOT-23-6 / DBV
U9	SN74LVC2G17	SN74LVC2G17DBVR	SOT-23-6 / DBV
F1	PTC 0.25 A	Specified generic	1206
R1	33 ohm	Specified generic	0603
R2	33 ohm	Specified generic	0603
R3	33 ohm	Specified generic	0603
R4	33 ohm	Specified generic	0603
R5	33 ohm	Specified generic	0603
R6	1 kohm	Specified generic	0603
R7	10 kohm	Specified generic	0603

Electrical ratings, capacitor derating and component-specific notes are in bom.csv. Generic passives are 1% resistors unless otherwise stated, with voltage and thermal ratings chosen for the specified rails. Confirm component availability and footprints during PCB/library release.

20. Bill of materials (20-38)

Ref	Value	Part / order specification	Package
R8	10 kohm	Specified generic	0603
R9	10 kohm	Specified generic	0603
R10	10 kohm	Specified generic	0603
R11	10 kohm	Specified generic	0603
R12	100 kohm	Specified generic	0603
R13	100 kohm	Specified generic	0603
R14	100 kohm	Specified generic	0603
R15	100 kohm	Specified generic	0603
R16	100 kohm	Specified generic	0603
R17	100 ohm	Specified generic	0603
R18	100 kohm	Specified generic	0603
SW1	RESET	Specified generic	Momentary normally open
C1	10 uF / 16 V	Specified generic	0805 X7R
C2	1 uF / 10 V	Specified generic	0603 X7R
C3	10 uF / 10 V	Specified generic	0805 X7R
C4	1 uF / 10 V	Specified generic	0603 X7R
C5	10 uF / 10 V	Specified generic	0805 X7R
C6	1 uF / 10 V	Specified generic	0603 X7R
C7	10 uF / 10 V	Specified generic	0805 X7R

Electrical ratings, capacitor derating and component-specific notes are in bom.csv. Generic passives are 1% resistors unless otherwise stated, with voltage and thermal ratings chosen for the specified rails. Confirm component availability and footprints during PCB/library release.

20. Bill of materials (39-55)

Ref	Value	Part / order specification	Package
C8	100 nF	Specified generic	0603 X7R
C9	100 nF	Specified generic	0603 X7R
C10	100 nF	Specified generic	0603 X7R
C11	1 nF C0G	Specified generic	0603 C0G
C12	1 nF C0G	Specified generic	0603 C0G
C13	1 nF C0G	Specified generic	0603 C0G
C14	100 nF	Specified generic	0603 X7R
C15	100 nF	Specified generic	0603 X7R
C16	100 nF	Specified generic	0603 X7R
C17	100 nF	Specified generic	0603 X7R
C18	100 nF	Specified generic	0603 X7R
C19	100 nF	Specified generic	0603 X7R
C20	1 uF	Specified generic	0603 X7R
C21	100 nF	Specified generic	0603 X7R
C22	100 nF	Specified generic	0603 X7R
C23	100 nF	Specified generic	0603 X7R
C24	100 nF	Specified generic	0603 X7R

Electrical ratings, capacitor derating and component-specific notes are in bom.csv. Generic passives are 1% resistors unless otherwise stated, with voltage and thermal ratings chosen for the specified rails. Confirm component availability and footprints during PCB/library release.

21. Reference-numeral register

Numerals	Meaning / first drawing
100,110	Acquisition device and digital interface; Figures 1-2
200,210-212	Trusted relay, normalization, canonical sample and order validation; Figures 1-3
220-224	Neuromorphic circuit, gate, selector, counter and spiking profiles; Figures 1-3
230-231	Count memory and frozen-frame readout; Figures 1 and 5
240-244	FHE/policy, privacy profile, encrypted representations and binding; Figures 1 and 11
250-270	Host, serial link, power and clock; Figure 2
300	Remote encrypted inference; Figures 1 and 13
401-406	Magnitude and refractory acceptance circuits; Figure 4
501-504	Address and saturating count update; Figure 5
601-602	Self-input and adjacent-neuron synapses; Figure 6
701-705	Neuron arithmetic and state selection; Figure 7
801-804	Time advance, decay and refractory state; Figure 8
901-905	Serial input, checking, controller and reply; Figure 9
1001-1005	Power sequence and configuration reset; Figure 10
1201-1205	Homomorphic linear-score arithmetic; Figure 12
1301-1305	Key separation and result decryption; Figure 13
1401-1406	Frame lifecycle and fault handling; Figure 14
1501-1505	Adaptive admission and routing; Figure 15
1601-1606	Optional training and deployment gate; Figure 16

Electrical designators U1, R1, C1 and J1 belong to the circuit/BOM namespace. Patent reference numerals are separate. U1 implements 220, 221, 223, 224 and their detailed circuits. M1 implements 210, 240 and 250. U2-U7 implement the power elements in Figure 10.

22. Sources and file guide

Authoritative sources were checked for pin assignments, voltage sequencing, supported device behavior and drawing conventions. The source repository commit for this package is c42d68225934a3bfa46a4bdd7573d035ad907b96. This is a design provenance record, not a statement about legal priority or public-disclosure dates.

ENER patent source, GitHub main

Raspberry Pi 4 Model B datasheet, release 1.1

Lattice iCE40 UltraPlus datasheet, FPGA-DS-02008 rev. 2.4

Lattice UltraPlus breakout-board schematic, Appendix A, rev. 1.2

Lattice iCE40 programming and configuration, section 13

Texas Instruments TLV755P datasheet, rev. D

Texas Instruments TPS3808 datasheet, rev. N

Texas Instruments SN74LVC2G17 datasheet, rev. N

SiTime SiT8008B datasheet, rev. 1.08

OpenFHE BFVms parameter-generation source

OpenFHE example parameter guidance

USPTO MPEP 608.02, drawing requirements

File / folder	Use
svg/	Editable vectors for every patent and circuit sheet
cad/	Legacy KiCad import project and cached symbols; import/ ERC not performed here
rtl/	Hardware source, Verilog, pin constraints and routed configuration artifacts
verification/	Simulation, synthesis, routing, connectivity and artifact review records
bom.csv / pin_net_map.csv	Complete physical component and pin tables
ENER_specification_with_reference_design.md	Original specification working copy with replaced drawing section and new F.14
drawing_descriptions.md	Matching sixteen-figure descriptions and numeral occurrence map